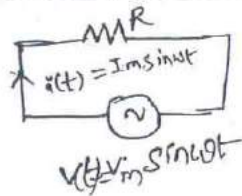


AC Source Across Resistor:-

13



$$i(t) = \frac{v(t)}{R}$$

$$i(t) = \frac{V_m \sin \omega t}{R} = \frac{V_m}{R} \sin \omega t$$

$$P(t) = I_m \sin \omega t$$

$$\begin{aligned} P(t) &= v(t) \cdot i(t) \\ &= V_m \sin \omega t \cdot I_m \sin \omega t \\ &= V_m I_m [\sin^2 \omega t] \\ &= \frac{V_m I_m}{2} [1 - \cos 2\omega t] \end{aligned}$$

$$P(t) = \frac{V_m I_m}{2} [1 - \cos 2\omega t] \rightarrow \text{Instantaneous power}$$

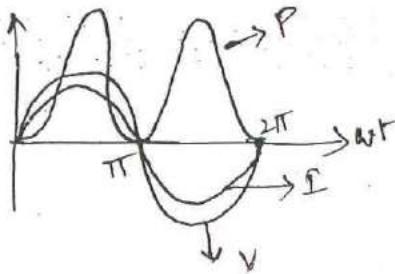
$$\text{Now Average power} = \frac{1}{2\pi} \int_0^{2\pi} P(t) \cdot d\omega t$$

Here 0 to 2π are taking. Since power is produced because of voltage & current

$$P_{avg} = \frac{1}{2\pi} \cdot \frac{V_m I_m}{2}$$

$$= \frac{V_m}{\sqrt{2}} \cdot \frac{I_m}{\sqrt{2}}$$

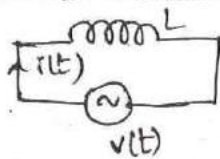
$$P_{avg} = V_{rms} I_{rms}$$



from this whenever voltage completed 1 cycle, then power completed 2 cycles. So $f = 50 \text{ Hz}$ (or) cycles/sec
 $f_{\text{power}} = 100 \text{ cycles/sec}$

(note:- Always frequency w.r. to voltage form (or) current waveform)

AC Source Across Inductor:-



$$v = L \frac{di}{dt}$$

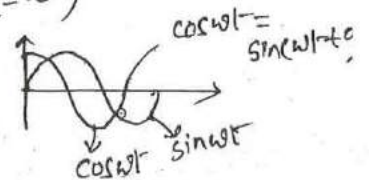
$$= L \cdot \frac{d(I_m \sin \omega t)}{dt}$$

$$v = \omega L I_m \cos \omega t$$

$$v(t) = X_L I_m \cos \omega t$$

$$v(t) = V_m \sin (\omega t + 90^\circ)$$

$$(\because X_L = \omega L)$$



$\therefore$ Instantaneous Power $P(t) = v(t) \cdot i(t)$

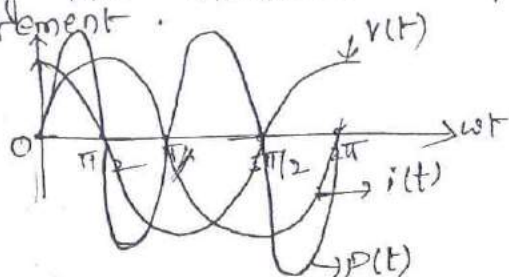
$$P(t) = V_m \cos \omega t \cdot I_m \sin \omega t$$

$$P(t) = \frac{V_m I_m}{2} \sin 2\omega t$$

$$P_{avg} = \frac{1}{2\pi} \int_0^{2\pi} P(t) d\omega t$$

$$= 0 \dots$$

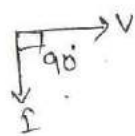
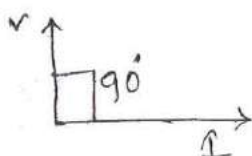
We know that Inductor never dissipates the energy. So by using this mathematical expression we can support that statement.



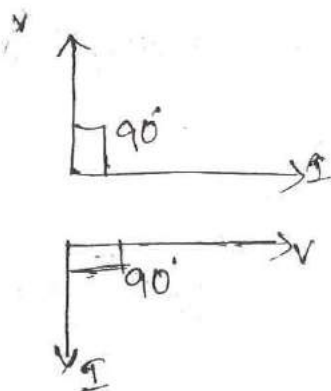
$$f = 50\text{Hz}$$

$$f_p = 100\text{Hz}$$

Note:- Always phasor diagrams are developed by using rms (values) or quantities.



In the +ve half cycle of the power wave Inductor takes energy from the source & in the -ve half cycle of the power wave Inductor delivers energy to the source. Thereby Net power taking from the source is equal to zero.



Note:- In a series circuit generally current is the reference.
In parallel " " voltage

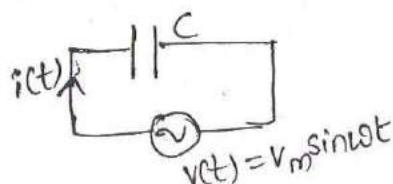
Note:-

$10 \angle 30^\circ$	→ vector
$\frac{10}{\sqrt{2}} \angle 30^\circ$	→ phasor
KCL, KVL possible...	

phasor → Two dimensional qty.

vector → Multi " "

AC source across capacitor:-



$$i = C \frac{dv}{dt}$$

$$= C \cdot \frac{d(V_m \sin \omega t)}{dt}$$

$$= \omega C \cdot V_m \cos \omega t$$

$$= \frac{V_m}{(\omega C)} \cos \omega t$$

* Direct addition ^{sub} is possible in phasor notation. Never possible in vector notation.
* phasor notation is utilized to represent only for voltage & current.

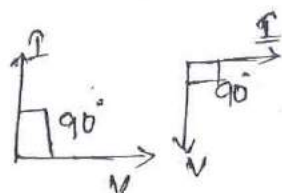
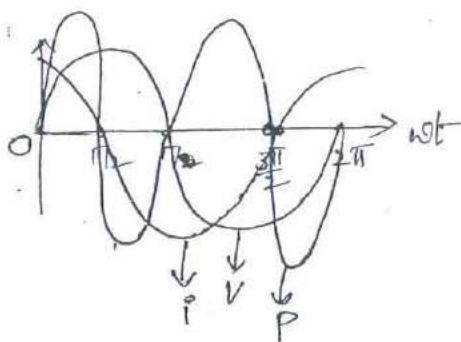
$$i(t) = I_m \cdot \sin(\omega t + 90^\circ)$$

Now

$$P(t) = v(t) \cdot i(t) \\ = v_m \sin \omega t \cdot I_m \cos \omega t$$

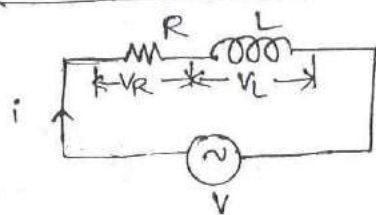
$$P(t) = \frac{v_m I_m}{2} \cdot \sin 2\omega t$$

$$P_{avg} = \frac{1}{2\pi} \int_0^{2\pi} P(t) d\omega t \\ = 0 \dots$$



$$f = 50 \text{ Hz} \\ f_p = 100 \text{ cycles/sec}$$

R-L Series circuit:-

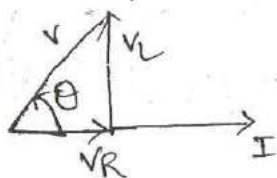


By KVL:-

$$V = V_R + V_L \\ V = V_R + j I X_L$$

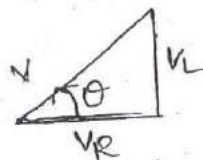
$$V = I R + j I X_L \\ = I [R + j X_L]$$

$$V = I Z \\ \text{where } Z = R + j X_L$$



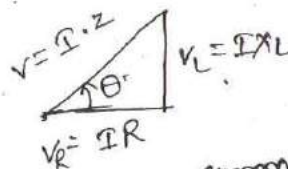
Phasor dgm

Vector dgm:-

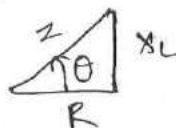


$$V = \sqrt{V_R^2 + V_L^2}$$

$$\theta = \tan^{-1} \left(\frac{V_L}{V_R} \right)$$

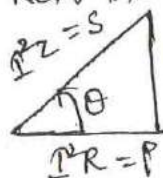


Here common term is current. By simplifying this we can obtain impedance vector dgm.



$$Z = \sqrt{R^2 + X_L^2} \\ \theta = \tan^{-1} \left(\frac{X_L}{R} \right)$$

Now the power triangle is given by



$$I^2 R = P$$

$$I^2 X_L = Q_L$$

$$S = \sqrt{P^2 + Q_L^2}$$

$$\theta = \tan^{-1} \left(\frac{Q_L}{P} \right)$$

corresponding to power triangle

P → Active Power (or) Real Power (or) True Power (or)
Average Power (or) effective Power

units → watts

$Q_L \rightarrow$ Inductive Reactive power unit : VAR
(volt ampere reactive)

$S \rightarrow$ Apparent power (or)
Complex Power units : VA
(volt ampere)

Instantaneous Power :-

$$\begin{aligned}
 p(t) &= v(t) \cdot i(t) \\
 &= V_m \sin(\omega t + \theta) I_m \sin \omega t \\
 &= V_m \sin(\omega t + \theta) \cdot I_m \sin \omega t
 \end{aligned}$$

Now

$$P_{avg} = \frac{1}{2\pi} \int_0^{2\pi} P(t) d\omega t$$
$$= \frac{1}{2\pi} \int_0^{2\pi} v_m I_m \sin(\omega t + \theta) \sin \omega t d\omega t$$

$$P_{avg} = V_{rms} \frac{I_{rms}}{2} \cos \theta$$

$$= \frac{V_{rms}}{\sqrt{2}} \frac{I_{rms}}{\sqrt{2}} \cos \theta$$

$$P_{avg} = V_{rms} I_{rms} \cos \theta$$

$$\text{Power factor} = \cos \theta = \frac{V_R}{V} = \frac{R}{Z} = \frac{P}{S} \quad (\text{lag})$$

for defining P.F. ^{nature (lag or lead)} for any A/P take voltage phasor as reference.
Why don't we take current as ref? — explanation is given below.

- 1) Suppose $R \Rightarrow I$, but V_{source} const. while defining P.F we have to take V constant vry. That's why voltage is taken as ref.
- 2) In the real time applications all are parallel circuits. So we are taking voltage as ref.

Note:- while defining nature of P.F for any combination of network voltage phasor is taken as refⁿ since

- 1) Source Voltage is constant.
- 2) In the real time system loads are connected in parallel.
(load voltage is constant)

- * Power factor angle indicates position of current phasor w.r.to voltage phasor.
- ** power factor defines contribution of reactive load in the total load of the system.
- * In the practical system always High p.f is desirable.

$$P = I^2 R = VI \cos \phi$$

$$Q_L = I^2 X_L = VI \sin \phi$$

$$S = I^2 Z = VI^* = V^* I$$

Explanation:-

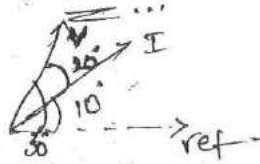
$$V = 10 \angle 30^\circ$$

$$I = 5 \angle 10^\circ$$

$$S = V \cdot I$$

$$= 10 \angle 30^\circ \cdot 5 \angle 10^\circ$$

$$= 50 \angle 40^\circ$$



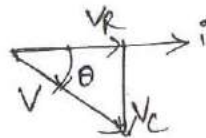
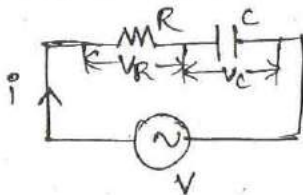
$$S = VI^*$$

$$= 10 \angle 30^\circ \cdot 5 \angle -10^\circ$$

$$= 50 \angle 20^\circ$$

from this the angle b/w V & $I = 20^\circ$ not 40° . So VI^* is absolutely correct.

RC series circuit:-



By KVL:-

$$V = V_R \angle 0^\circ + V_C \angle -90^\circ$$

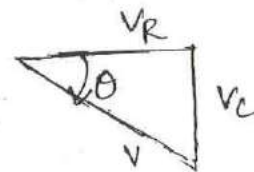
$$= V_R - jV_C$$

$$= IR - jIX_C$$

$$= (R - jX_C) I$$

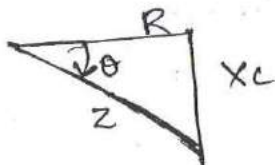
$$V = I \cdot Z$$

where $Z = R - jX_C$



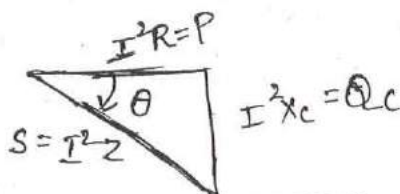
$$V = \sqrt{V_R^2 + V_C^2}$$

$$\theta = \tan^{-1} \left(\frac{-V_C}{V_R} \right)$$



$$Z = \sqrt{R^2 + X_C^2}$$

$$\theta = \tan^{-1} \left(-\frac{X_C}{R} \right)$$



$$S = \sqrt{P^2 + Q_C^2}$$

$$\theta = \tan^{-1} \left(-\frac{Q_C}{P} \right)$$

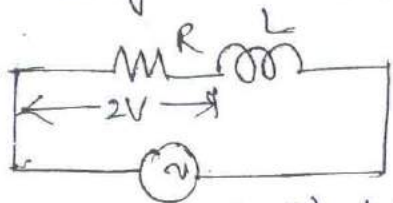
Power factor = $\cos \theta$

$= \frac{V_R}{V}$

$= \frac{R}{Z}$

$= \frac{P}{S}$ (true)

(Pb) find voltage across inductor from the circuit shown?

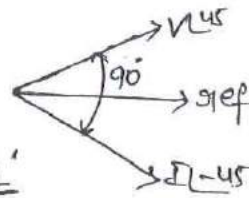


(Pb) Find ckt element for a given voltage & current eqns?

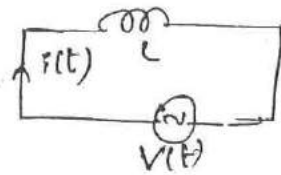
$$v(t) = 9 \sin(t + 45^\circ)$$

$$i(t) = 3 \sin(t - 45^\circ)$$

Sol:-



so it is 'L'



$$X_L = \frac{V}{I}$$

$$= \frac{9/12}{3/12}$$

$$= 3 \Omega$$

$$\omega L = 3$$

$$L = \frac{3}{\omega}$$

$$= \frac{3}{2\pi} \frac{3}{1}$$

$$L = 8 \text{ H}$$

Sol:-

$$V_{rms} = \sqrt{V_R^2 + V_L^2}$$

$$\frac{10}{\sqrt{2}} = \sqrt{4^2 + V_L^2}$$

$$\Rightarrow \frac{100}{2} = 4 + V_L^2$$

$$\Rightarrow V_L^2 = 46$$

$$\Rightarrow V_L = \sqrt{46} \text{ volts}$$

(Pb)

$$v(t) = 9 \sin(t + 30^\circ)$$

$$i(t) = 3 \sin(2t + 60^\circ)$$

Sol:-

NOTE:- By using above eqn it is not possible to design the N/w since freq of voltage & current are unequal.

(Pb)

$$v(t) = 10 \sin t$$

$$i(t) = 5 \sin(t + 60^\circ) + 3 \sin(3t + 60^\circ)$$

Sol:-

while calculating the power we have to concentrate on equal freq.

2nd harmonic voltage = 0, 3rd harmonic current = 0 $\Rightarrow$ 3rd har|| power = 0.

$$P = VI \cos \phi$$

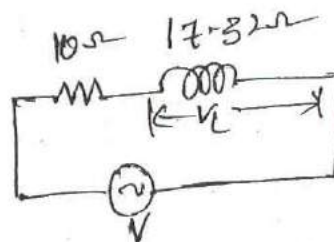
$$= \frac{10}{\sqrt{2}} \text{ rms} \cdot \frac{5}{\sqrt{2}} \cos(60^\circ)$$

$$= \frac{50}{2} \sin \phi$$

$$= \left(\frac{10}{\sqrt{2}}\right) \left(\frac{5}{\sqrt{2}}\right) \cos(60^\circ)$$

$$= 12.5 \text{ W}$$

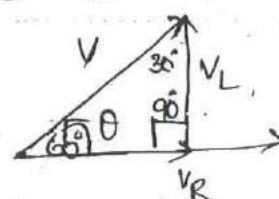
(Pb)



find angle made by source voltage w.r to V_L ?

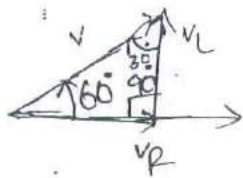
(a) 30° (b) -30° (c) 60° (d) -60°

Sol:-



$$\theta = \tan^{-1}\left(\frac{X_L}{R}\right)$$

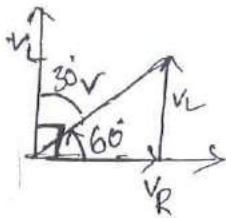
$$= \tan^{-1}\left(\frac{17.32}{10}\right) = 60^\circ$$



Ans: -30° (since CW)

means w.r.t to I voltage (if moving) is angle.

(OR)



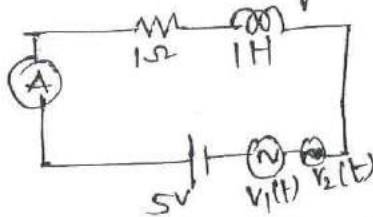
w.r.t to V_L , V is lagging by 30° .

so Ang = -30° ...

Note: When phasor rotates in the CW direction sign for the angle is -ve.

(Pb)

find Ammeter reading & P.F. of the ckt shown.



$$v_1(t) = 10 \sin 2t$$

$$v_2(t) = 10\sqrt{5} \sin 2t$$

Sol: phasor concept is applicable only for equal frequency. When different frequency are exist we have to use superposition theorem.

V_1 alone:-

$$X_L = \omega L = 1 \Omega$$

$$Z_1 = \sqrt{R^2 + X_L^2} = \sqrt{1+1} = \sqrt{2}$$

$$i_1 = \frac{V_1}{Z_1} = \frac{10/\sqrt{2}}{\sqrt{2}} = \frac{10}{2} = 5A$$

V_2 alone

$$X_L = \omega L = 2$$

$$Z_2 = \sqrt{R^2 + X_L^2}$$

$$= \sqrt{1^2 + 2^2} = \sqrt{5}$$

$$i_2 = \frac{10\sqrt{5}/\sqrt{5}}{\sqrt{5}} = \frac{10}{\sqrt{2}} A \dots$$

V_3 (D.C)

$$\omega = 0$$

$$X_L = 0$$

$$Z = R$$

$$i_3 = \frac{V_3}{R} = \frac{5}{1} = 5A$$

When the given N/W is having multiple frequency

$$* i = \sqrt{i_1^2 + i_2^2 + i_3^2} = \sqrt{5^2 + \left(\frac{10}{\sqrt{2}}\right)^2 + 5^2} = 10A \dots$$

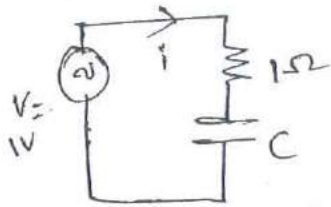
P.F.:- when diff. frequency exist the P.F. can be getting only from the power Triangle. Not use impedance triangle.

$$\cos \theta = \frac{P}{S} = \frac{i^2 R}{i^2 Z} \quad \frac{i^2 R}{V i} = \frac{i R}{V}$$

$$\text{NOW } V = \sqrt{V_1^2 + V_2^2 + V_3^2} = \sqrt{\left(\frac{10}{\sqrt{2}}\right)^2 + \left(\frac{10\sqrt{5}}{\sqrt{2}}\right)^2 + 5^2}$$

$$\therefore P.F. = \frac{(10)(1)}{\sqrt{275}} \approx 0.55$$

- (pb) In the ckt shown, power dissipation is 500 mW, find phase displacement b/w source voltage & source current?



Note:- If $V=1V$ is given by default take it as 1mV.

Sol:-

$$P = i^2 R = \left(\frac{V}{Z}\right)^2 \cdot R$$

$$= \left[\frac{V}{\sqrt{R^2 + X_C^2}}\right]^2 \cdot R$$

$$500 \times 10^{-3} = \left(\frac{1}{\sqrt{1^2 + X_C^2}}\right)^2 (1) \Rightarrow X_C = 1$$

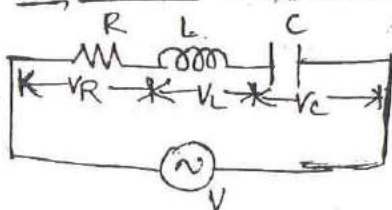
$$\therefore Z = R - jX_C = 1 - j(1)$$

$$\theta = \tan^{-1}\left(-\frac{X_C}{R}\right) = \tan^{-1}\left(-\frac{1}{1}\right) = -45^\circ \dots$$

$$i = \frac{V}{Z} = \frac{1 \angle 0^\circ}{1 \angle -45^\circ} = \frac{1}{1} \angle 45^\circ \dots$$

Note:- Always impedance angle & P.F angle both are having same magnitude with opposite sign.

RLC series circuit :-

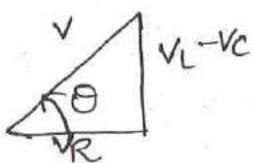


By KVL :-

$$V = V_R \angle 0^\circ + V_L \angle 90^\circ + V_C \angle -90^\circ$$

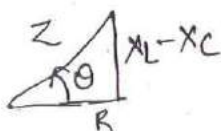
$$IZ = IR + jI X_L + (-j)IX_C$$

$$Z = R + j(X_L - X_C)$$



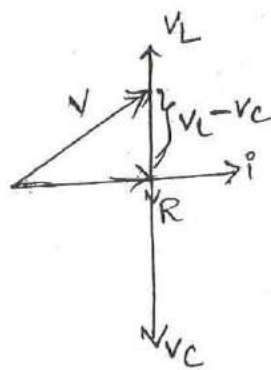
$$V = \sqrt{V_R^2 + (V_L - V_C)^2}$$

$$\theta = \tan^{-1}\left(\frac{V_L - V_C}{V_R}\right)$$



$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

$$\theta = \tan^{-1}\left(\frac{X_L - X_C}{R}\right)$$



(i is ref. since series ckt)

$$V_L > V_C$$

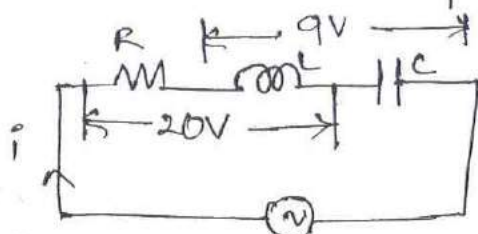
$$\begin{array}{c} I^2 Z = S \\ \uparrow \theta \\ I^2 R = P \end{array} \quad I^2 (X_L - X_C) = Q_L - Q_C$$

w.k.T
 $S = \sqrt{P^2 + Q^2}$
 But here

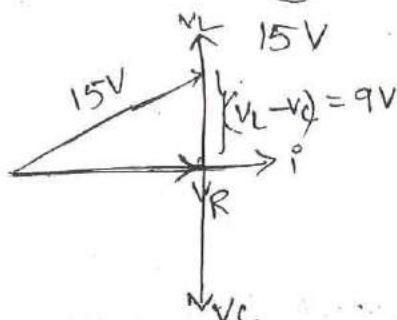
$$S = \sqrt{P^2 + (Q_L - Q_C)^2} //$$

$$\theta = \tan^{-1} \left(\frac{Q_L - Q_C}{P} \right) //$$

(pb) Find voltage across capacitor of the circuit shown?
 (a) 9 (b) 7 (c) 7 or 25 (d) none



Sol:-



Assum $V_L > V_C$

$$V_R = \sqrt{(15)^2 - 9^2} = 12V$$

$$\sqrt{V_R^2 + V_L^2} = 20V \Rightarrow V_L = 16V$$

$$V_L - V_C = 9V \Rightarrow 16 - V_C = 9$$

$$\Rightarrow V_C = 7V //$$

from the given data $V_L - V_C = 9$ (or) $V_C - V_L = 9$

$$V_L - V_C = \pm 9$$

$$V_L - V_C = 9$$

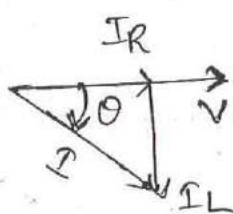
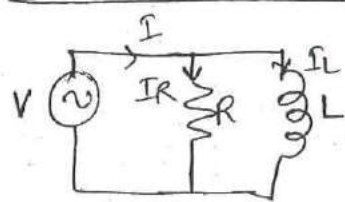
$$V_L - V_C = -9$$

$$16 - V_C = -9$$

$$V_C = 25V$$

we don't know which one is greater. That's why two possibilities are occurs.

Parallel RL circuit:-



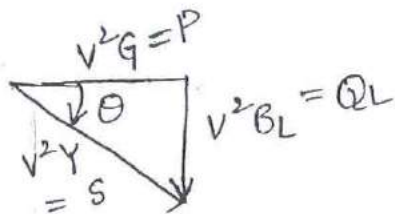
By KCL:-

$$I = I_R \angle 0 + I_L \angle -90$$

$$I = I_R - j I_L \Rightarrow \frac{V}{Z} = \frac{V}{R} - j \frac{V}{X_L}$$

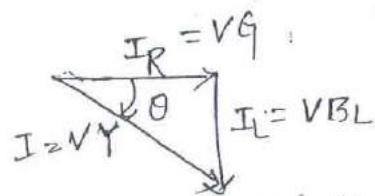
$$V \cdot Y = V G - j V B_L$$

$$\Rightarrow Y = G - j B_L$$



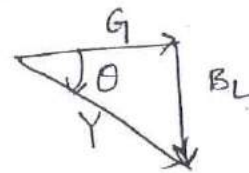
$$S = \sqrt{P^2 + Q_L^2}$$

$$\theta = \tan^{-1} \left(-\frac{Q_L}{P} \right)$$



Common term is the voltage
 $\Downarrow$

Admittance triangle:



$$Y = \sqrt{G^2 + B_L^2}$$

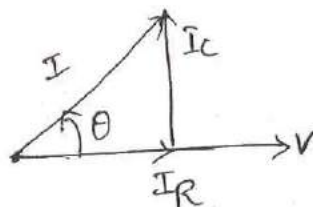
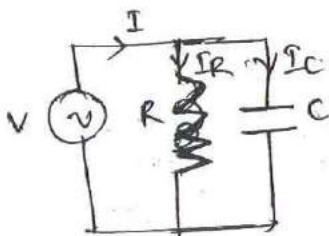
$$\theta = \tan^{-1} \left(-\frac{B_L}{G} \right)$$

$\equiv \dots$

$$\text{Power factor} = \cos \theta = \frac{I_R}{I} = \frac{G}{Y} = \frac{P}{S} \quad (\text{lag})$$

(note:- for P.F. always voltage is the reference)

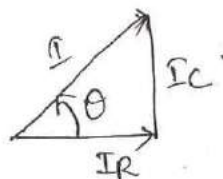
RC Parallel Circuit:-



$$I = I_R + I_C$$

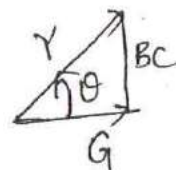
$$\frac{V}{Z} = \frac{V}{R} + j \frac{V}{X_C}$$

$$Y = G + j B_C$$



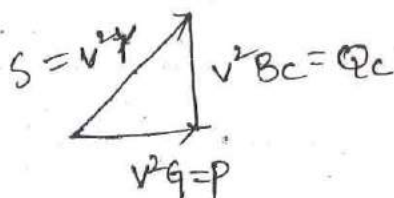
$$I = \sqrt{I_R^2 + I_C^2}$$

$$\theta = \tan^{-1} \left(\frac{I_C}{I_R} \right)$$



$$Y = \sqrt{G^2 + B_C^2}$$

$$\theta = \tan^{-1} \left(\frac{B_C}{G} \right)$$

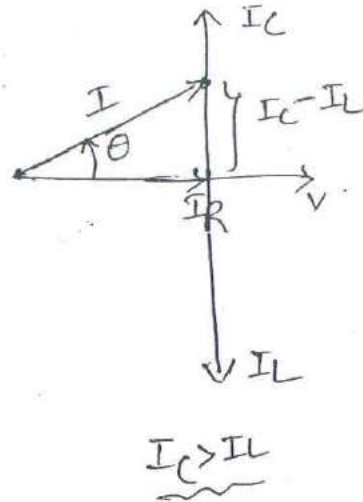
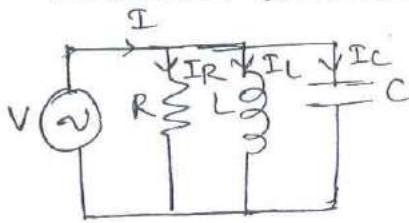


$$S = \sqrt{P^2 + Q_C^2}$$

$$\theta = \tan^{-1} \left(\frac{Q_C}{P} \right) \dots$$

$$\text{P.F.} = \cos \theta = \frac{I_R}{I} = \frac{G}{Y} = \frac{P}{S} \quad (\text{lead})$$

RLC parallel circuit:-

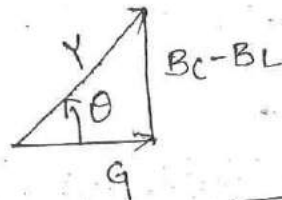
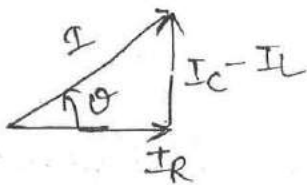


$$I = I_R \angle 0 + I_L \angle -90 + I_C \angle 90$$

$$\frac{V}{Z} = \frac{V}{R} + \frac{V}{X_L} (-j) + j \frac{V}{X_C}$$

$$VY = V G + -j V B_L + j V B_C$$

$$Y = G + j(B_C - B_L)$$



$$I = \sqrt{I_R^2 + (I_C - I_L)^2}$$

$$\theta = \tan^{-1} \left(\frac{I_C - I_L}{I_R} \right)$$

$$Y = \sqrt{G^2 + (B_C - B_L)^2}$$

$$\theta = \tan^{-1} \left(\frac{B_C - B_L}{G} \right)$$

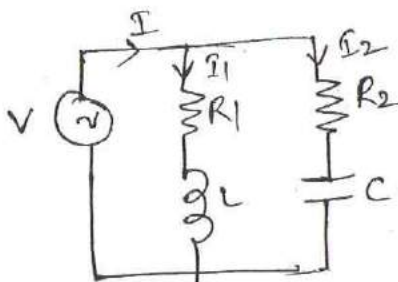
$$S = \sqrt{P^2 + (Q_C - Q_L)^2}$$

$v^2 G = P$

$$S = \sqrt{P^2 + (Q_C - Q_L)^2}$$

$$\theta = \tan^{-1} \left(\frac{Q_C - Q_L}{P} \right)$$

(Pb)



find conductance of branch 1 & 2?

sol:

$$I_1 = \frac{V}{R_1 + jX_L}$$

$$I_1 = \frac{V [R_1 - jX_L]}{(R_1 + jX_L)(R_1 - jX_L)} = \frac{V R_1}{R_1^2 + X_L^2} - \frac{j X_L V}{R_1^2 + X_L^2}$$

w.r.t $I = V \cdot G - j V \cdot B_L \Rightarrow G = \frac{R_1}{R_1^2 + X_L^2}$

$$\text{Now } I_2 = \frac{V}{R - jX_C}$$

$$= \frac{V}{(R - jX_C)} \times \frac{(R + jX_C)}{(R + jX_C)}$$

$$I_2 = V \left[\frac{R}{R^2 + X_C^2} + j \frac{X_C}{R^2 + X_C^2} \right]$$

$$\Rightarrow G = \frac{R}{R^2 + X_C^2}$$

.....

$$\text{Now } I = I_1 + I_2$$

$$VY = V[Y_1] + VY_2$$

$$\Rightarrow \boxed{Y = Y_1 + Y_2}$$

for parallel circuit....

$$Y = Y_1 + Y_2 + \dots + Y_n$$

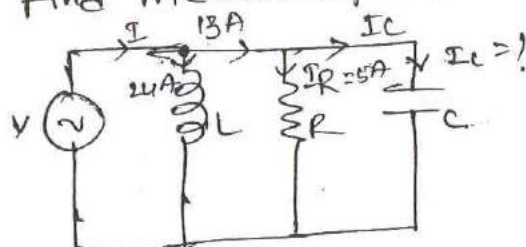
$$= (G_1 - jB_L) + (G_2 + jB_C)$$

$$\underline{Y = (G + G_2) + j(B_C - B_L)}$$

.....

(Pb)

find the value of I_C & I ?



Sol:-

$$13 = \sqrt{I_C^2 + 5^2}$$

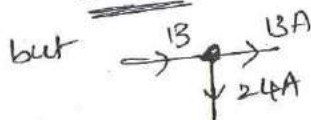
$$\Rightarrow \underline{I_C = 12A}$$

.....

$$\boxed{I = \sqrt{(I_R)^2 + (I_L - I_C)^2}}$$

At node:

$$\underline{I = 13}$$

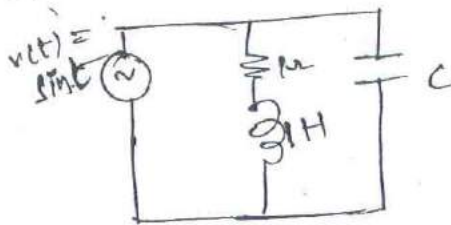


KCL not satisfied!

AC $\rightarrow$ KCL, KVL $\rightarrow$ phasor sum
 DC $\rightarrow$ KCL, KVL $\rightarrow$ Arithmetic sum

So KCL is satisfied.

(Pb) find the value of 'C' when the P.F of the ckt is 0.8 lag



Sol:- for branch 1
 $X_L = \omega L = 1 \Omega$

$$G = \frac{R_1}{R_1^2 + X_L^2} = \frac{1}{1+1} = \frac{1}{2}$$

$$B_L = \frac{X_L}{R_1^2 + X_L^2} = \frac{1}{1+1} = \frac{1}{2}$$

for branch 2:-

$$Y_2 = jB_C$$

$$= j\omega C$$

$$Y_2 = j(1)C$$

$$Y_1 = G_1 - jB_L = \frac{1}{2} - j\frac{1}{2}$$

$$\therefore Y_{eq} = Y_1 + Y_2 = \frac{1}{2} + \left[-j\frac{1}{2}\right] + jC$$

$$Y_{eq} = \frac{1}{2} + j\left(C - \frac{1}{2}\right)$$

$$\cos \theta = \frac{G}{|Y_{eq}|} \Rightarrow \frac{\frac{1}{2}}{\sqrt{\left(\frac{1}{2}\right)^2 + \left(C - \frac{1}{2}\right)^2}} = 0.8$$

$$\Rightarrow C = \frac{1}{8} \text{ (or) } \frac{7}{8}$$

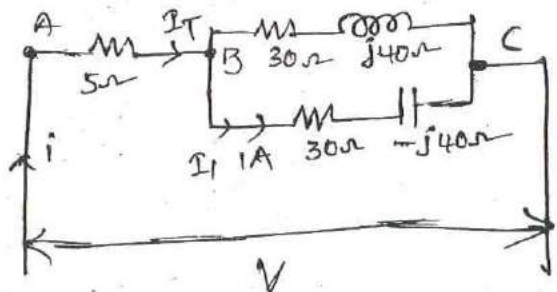
To satisfy the lag P.F condition $\Rightarrow B_L > B_C$
 $\frac{1}{2} > C$

$\Rightarrow C = \frac{1}{8}$ satisfy this condition.

Note:- for exth of the given problem
 have to take $\frac{1}{2} < C \Rightarrow C = \frac{7}{8}$

(Pb)

find the value of V of the ckt shown?



- (a) 50 (b) 55
 (c) 56 (d) 60

Sol:-

$$I_1 = I_T \times \frac{[30 + j40]}{[30 + j40] + [30 - j40]}$$

$$1 = I_T \times \frac{[30 + j40]}{60} \Rightarrow I_T = \frac{60}{50 \tan^{-1}(4/3)}$$

$$I_T = 1.2 \angle \tan^{-1}(-4/3)$$

$$V_{AB} = (1.2 \angle \tan^{-1}(-4/3)) \times 5$$

$$= 6 \angle \tan^{-1}(-4/3)$$

$$V_{BC} = 1 \angle 30^\circ - j40^\circ$$

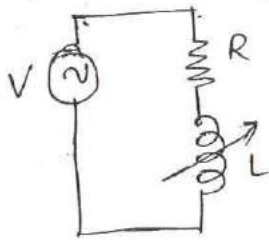
$$= 50 \angle \tan^{-1}(-4/3)$$

$$\therefore V_{AC} = V = V_{AB} + V_{BC}$$

$$= 50 \angle \tan^{-1}(-4/3) + 50 \angle \tan^{-1}(-4/3)$$

$$= 56 \angle \tan^{-1}(-4/3)$$

Locus Diagram :-



→ for Analysis & designing of the ckt's.

→ Practical Application is "filters".

→ for ^{finding} Performance of m/c, also we are using this.

→ Locus dgm can be drawn by selecting any one of the element & varying (or) All the elements are constant & frequency vary.

* Locus diagrams are useful for Analysis & designing of the ckt.

Ex: 1) Filters.

2) Induction motor (Circle diagram)

3) Synchronous machine (Excitation & Power circle).

* The path traced by Terminals of the current vectors by varying either any one of the circuit elements or by varying frequency is called as "current locus".

$$X_L = 2\pi f L$$

for pure resistive ckt $X_L = 0$

$$\Rightarrow Z = R$$

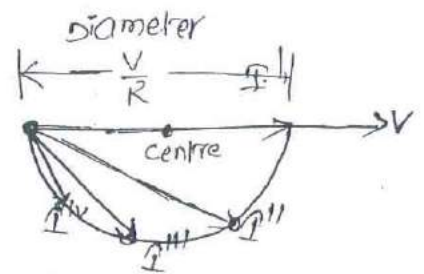
$$I = \frac{V}{R} ; \theta = 0^\circ$$

$$X_L = 2\pi fL$$

$$X_L \uparrow \Rightarrow Z \uparrow \Rightarrow I \downarrow ; \theta = \tan^{-1}\left(\frac{X_L}{R}\right)$$

$$X_L \uparrow \uparrow \Rightarrow Z \uparrow \uparrow \Rightarrow I \downarrow \downarrow ; \theta \uparrow \uparrow$$

$$X_L = \infty \Rightarrow Z = \infty \Rightarrow I = 0 ; \theta = 90^\circ$$



$$\text{Diameter} = \frac{V}{R}$$

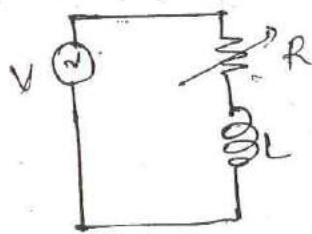
$$\text{Centre} = \left(\frac{V}{2R}, 0\right)$$

for the same N/w :

$R = \text{constant}$
 $L = \text{constant}$
 $f \text{ vary}$ } $\Rightarrow$ current locus diagram is same.

in the above N/w by keeping all the elements constant & by varying freqⁿ also, same shape of the current locus diagram is obtained.

Develop a current locus of the ckt shown.



$$R = 0 ;$$

$$\Rightarrow Z = X_L$$

$$I = \frac{V}{X_L}$$

$$\theta = 90^\circ$$

$$R \uparrow$$

$$Z \uparrow$$

$$I \downarrow$$

$$\theta = \tan^{-1}\left(\frac{X_L}{R}\right)$$

$$\theta \downarrow$$

$$R \uparrow \uparrow$$

$$Z \uparrow \uparrow$$

$$I \downarrow \downarrow$$

$$\theta \downarrow \downarrow$$

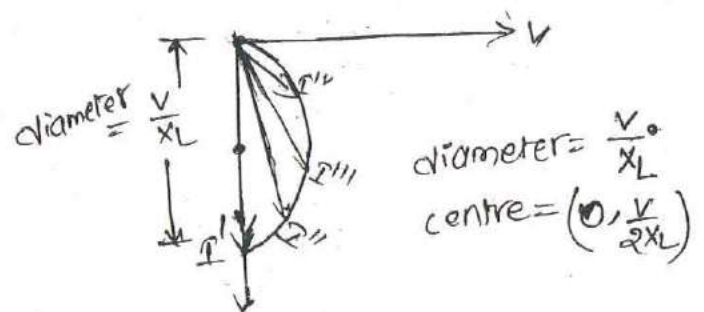
$$R = \infty$$

$$Z = \infty$$

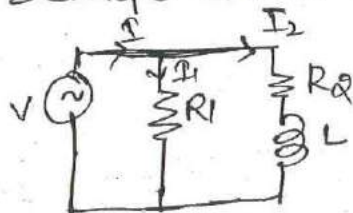
$$I = 0$$

note :-

from this we can observe that, Diameter is always w.r.to constant element & also Diameter is the initial current.



Develop Current locus of I_1 & I_2 of the ckt shown.



$$R_1 \approx 0.1$$

$$R_1 \neq 0$$

$$\text{If } R_1 = 0 \Rightarrow \text{S.C}$$

$$\Rightarrow \text{Eqⁿ Impedance} = 0$$

$$\Rightarrow I = \infty$$

So we are taking small resⁿ.

$$I_s = \frac{V_s}{R_1}$$

$$\theta = 0$$

$$I_T = I_1 + I_2 = I$$

$$R_1 \uparrow$$

$$Z_1 \uparrow$$

$$I_1 \downarrow$$

$$\theta_1 = 0$$

$$I_2 = \frac{V}{R_2 + jX_L} \Rightarrow I_2 = \text{constant}$$

$$I = I_1 + I_2$$

$$R_1 \uparrow \uparrow$$

$$Z_1 \uparrow \uparrow$$

$$\theta = 0$$

$$I_1 \downarrow \downarrow$$

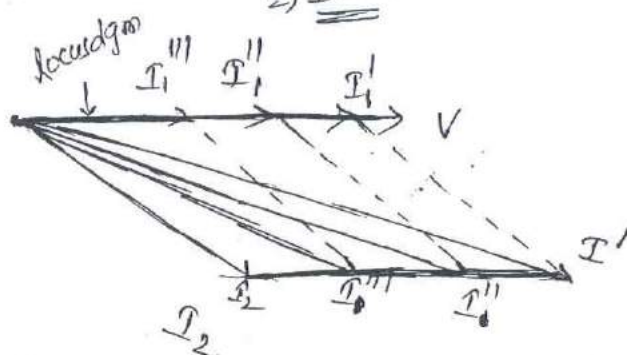
$$R_1 = \infty$$

$$Z_1 = \infty$$

$$\theta_1 = 0$$

$$I_1 = 0$$

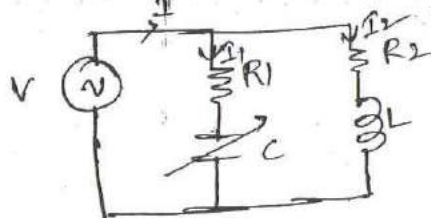
$$\Rightarrow I = I_2$$



note :- 1) when P.F. angle vary means the shape of the current locus is semi circle.

2) P.F. angle constant " " straight line.

Develop current locus of I_1 & I of the crt shown.



$$X_C = \frac{1}{2\pi f C}$$

$$C \approx \infty \Rightarrow X_C \approx 0$$

$$\Rightarrow Z_1 = R_1$$

$$I_1 = \frac{V}{R_1}$$

$$\theta_1 = 0$$

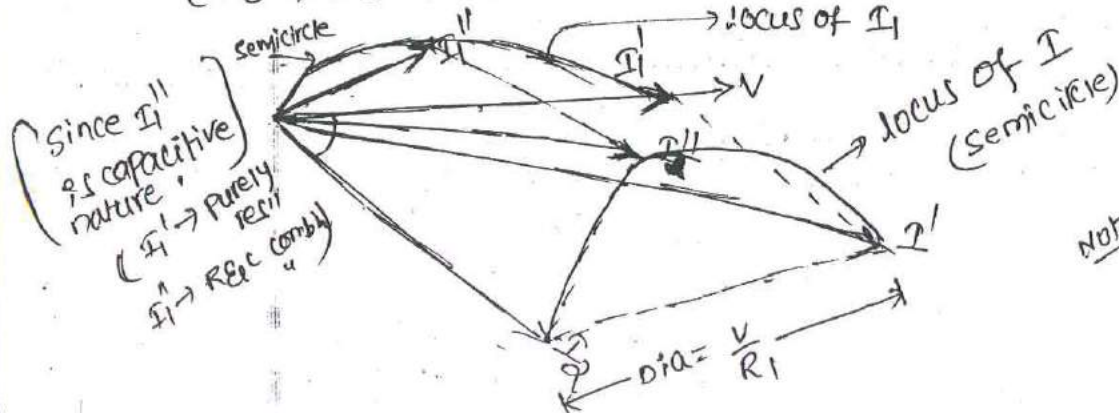
$$I = I_1 + I_2$$

$$C \downarrow \Rightarrow X_C \uparrow \Rightarrow Z_1 \uparrow \Rightarrow I_1 \downarrow \Rightarrow I \downarrow ; \theta = \tan^{-1}\left(-\frac{X_C}{R}\right) \uparrow$$

we are not focus on sign, just focus on magnitude.

$$C \downarrow \downarrow \Rightarrow X_C \uparrow \uparrow \Rightarrow I_1 \downarrow \downarrow ; \theta_1 \uparrow \uparrow$$

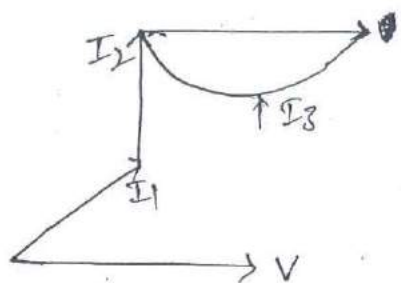
$$C = 0 \Rightarrow X_C = \infty \Rightarrow I_1 = 0 \Rightarrow I = I_2$$



Since I_1 is capacitive nature, $I_1 \rightarrow$ purely resistive $I_1 \rightarrow$ R & C comb.

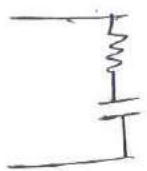
note: Here θ is vary so we got semi circle.

(Pb) Design a circuit for the given Current locus diagram.



Sol:- It is a parallel ckt.
No. of branches = 3.

w.r.to $V \Rightarrow I_1$ lead by 0° so
RC N/W.

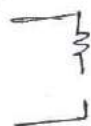


w.r.to $V \Rightarrow I_2$ lead by 90° so 'C'



Dia $\Rightarrow$ Anticlockwise current & also known
Dia " always due to is initial
Current.

Dia $\Rightarrow \frac{V}{R_1}$ for the given semi
circle.



when ever inductor
is the variable then
semicircle is downward
" " capacitor "
" clockwise "

so it is an inductor



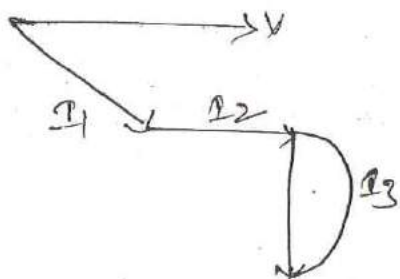
$\therefore$ final dgm is:



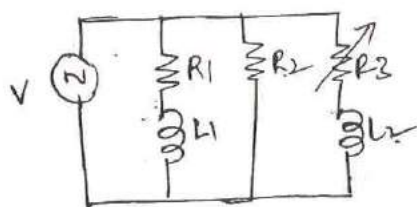
Note:-

Diameter of semicircle indicate
Anticlock current & it is obtained
by constant element.

(Pb) Design a ckt for the given
locus dgm.



Sol:-

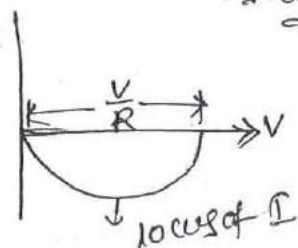
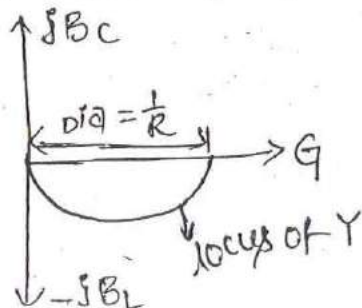


Note:- semicircle can be get only
for RC (or) RL combination.
we never get semicircle for
LC combination

Note:- $I = VY$

$I \propto Y$ for admittance also same.

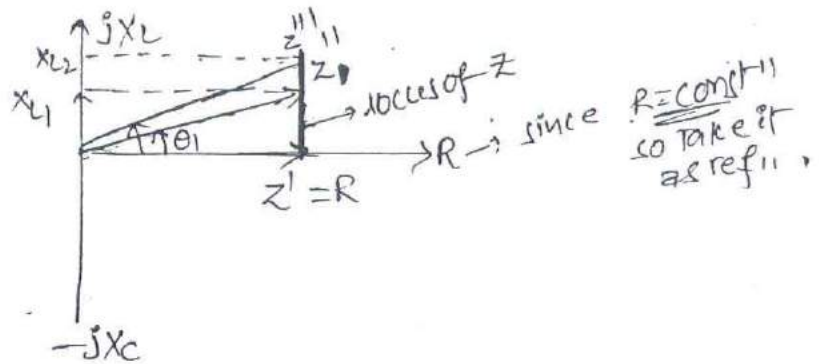
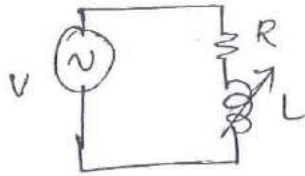
so the Current locus dgm & admittance locus dgm both are
same.



an current locus
dgm
voltages
are
included.
both
locus dgm
are same.

an Admⁿ locus dgm voltages not included,
that's why we $\Rightarrow \frac{1}{R}$.

Draw the Impedance locus diagram of the ckt shown.



$$X_L = 0$$

$$\Rightarrow Z = R$$

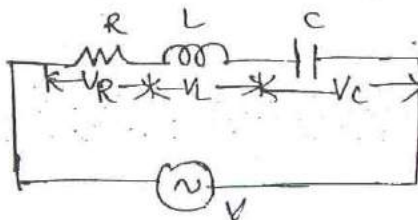
$$\theta = 0$$

$$X_L \uparrow$$

$$Z \uparrow$$

$$\theta = \tan^{-1}\left(\frac{X_L}{R}\right)$$

Resonance



- * For occurrence of resonance two energies are required.
- * Based on connection
 - series resonance.
 - parallel resonance.

* In this series resonance ckt current i is taken as reference.

* To satisfy the definition we take $V_L = V_C$.

* The freq. at which $X_L = X_C$ is called "resonant freq.".

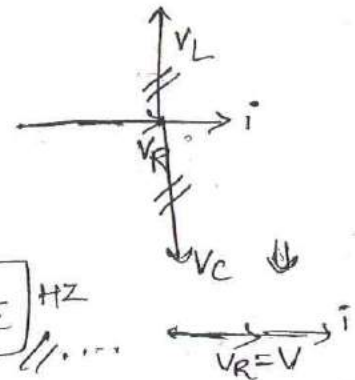
$$V_L = V_C$$

$$i X_L = i X_C$$

$$i \omega L = i \cdot \frac{1}{\omega C}$$

$$\Rightarrow \omega_0 = \frac{1}{\sqrt{LC}} \text{ rad/sec}$$

$$f_0 = \frac{1}{2\pi\sqrt{LC}} \text{ Hz}$$



For occurrence of the resonance in any system two energies are required. An RLC ckt inductor is having energy in the form of magtic field & capacitor is having energy in the form of electric field, when these two energies are present at particular freq. wide variations are present in the system is called as "resonance".

The ckt is said to be resonance when source current is inphase with source voltage.

The freq. at which $X_L = X_C$ is called as "Resonant frequency".

The Resonant freqⁿ indicates the state at which energy transformation is done b/w inductor & capacitor.

$$1. Z = R + j(X_L - X_C) \quad \underbrace{}_{=0}$$

$$\Rightarrow Z_{min} = R$$

$$2. I = \frac{V}{Z} \Rightarrow I_{max} = \frac{V}{Z_{min}} = \frac{V}{R}$$

$$3. \cos \theta = 1$$

$$4. V_R = V$$

$$5. \text{Net reactive voltage} = 0$$

6. voltage across inductor (or) voltage across capacitor is greater than source voltage. This phenomena is called as "voltage magnification".

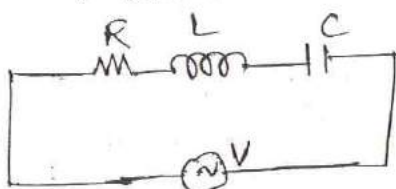
7. Applications:-

1) In the designing of oscillators.

2) filters. (BPF & BRF we use resonant phenomena).

3) Tuning circuits

4) Induction heating $\rightarrow$ (electrical application).



$$\begin{cases} \rightarrow V_C = IX_C \\ \rightarrow V_L = IX_L \\ \rightarrow V_R = IR \end{cases}$$

Variation of voltage across capacitor & voltage across inductor w.r.t. to freqⁿ:-

$$V_C = IX_C$$

$$X_L = 2\pi fL$$

$$X_C = \frac{1}{2\pi fC}$$

$$f=0$$

$$X_L=0$$

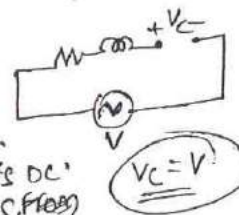
$$X_C=\infty$$

$$|X_L - X_C| = \infty$$

$$Z = \infty$$

$$I = 0$$

$$\Rightarrow V_C = V$$



In other words if $f=0 \Rightarrow$ source is DC.
As time $\uparrow \Rightarrow$ capacitor charges, freqⁿ $\rightarrow$ 0.
At particular instant $V_C = V$.

$$f \uparrow \Rightarrow X_L \uparrow ; X_C \downarrow ; X = |X_L - X_C| \downarrow \downarrow ; Z \downarrow \downarrow ; I \uparrow \uparrow$$

(100%) $\rightarrow$ increases (100%) $\rightarrow$ reduces

$V = IX_C \Rightarrow$ Here increasing current is dominant then decreasing X_C . So $V_C \uparrow$.

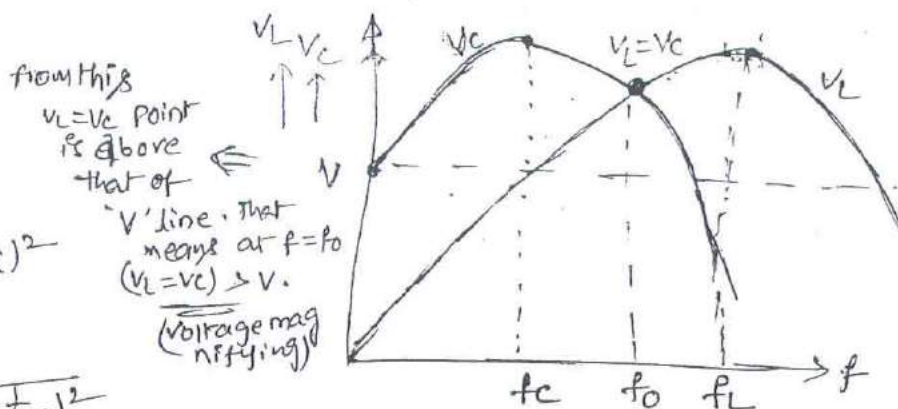
$$f \uparrow \uparrow \Rightarrow X_L \uparrow \uparrow ; X_C \downarrow \downarrow ; X = |X_L - X_C| \uparrow \uparrow ; Z = (R + j(X_L - X_C)) \uparrow \uparrow ; I \downarrow \downarrow ; V_C \downarrow$$

$$V_C = IX_C$$

$$= \frac{V}{Z} \cdot X_C$$

$$= \frac{V \cdot (X_C)}{\sqrt{R^2 + (X_L - X_C)^2}}$$

$$V_C = \frac{V X_C}{\sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2}}$$



NOW for getting ~~resonant freq~~ the frequ at which V_C is max, (i.e f_c)

$$\frac{dV_C}{d\omega} = 0 \Rightarrow \boxed{f_c = \frac{1}{2\pi\sqrt{LC}} \sqrt{1 - \left(\frac{R^2 C}{2L}\right)}}$$

NOW

$$V_L = IX_L$$

$$X_L = 2\pi fL$$

$$X_C = \frac{1}{2\pi fC}$$

$$f=0$$

$$X_L=0$$

$$X_C=\infty$$

$$X = |X_L - X_C| = \infty$$

$$Z = \infty; I = 0$$

$$\boxed{V_L = 0} *$$

$$f \uparrow, X_L \uparrow, X_C \downarrow, X = |X_L - X_C| \downarrow; Z \downarrow$$

(10Ω) (100Ω) (90Ω)

$$I \uparrow \uparrow, V_L \uparrow$$

$$\therefore I \uparrow, X_L \uparrow$$

$$f \uparrow \uparrow, X_L \uparrow \uparrow, X_C \downarrow \downarrow; Z = (R + j(X_L - X_C)) \uparrow \uparrow$$

$$I \downarrow \downarrow \therefore V_L \downarrow$$

for getting f_L :-

$$V_L = IX_L$$

$$V_L = \frac{V \cdot \omega L}{\sqrt{R^2 + (X_L - X_C)^2}}$$

$$V_L = \frac{V \omega L}{\sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2}} \rightarrow \textcircled{2}$$

NOW differentiate eqn ② with ω & equal to zero, we get

$$\frac{dV_L}{d\omega} = 0 \Rightarrow \boxed{f_L = \frac{1}{2\pi\sqrt{LC}} \sqrt{\frac{1 - \frac{R^2 C}{2L}}{1 - \left(\frac{R^2 C}{2L}\right)}}}$$

Q-Factor :-

Definition:- Quality factor is the ratio of maximum energy stored in the circuit to power dissipation per cycle.

$$W_g = \frac{1}{2} L i^2 + \frac{1}{2} C v^2$$

$$= \frac{1}{2} L (I_m \sin \omega t)^2 + \frac{1}{2} C \left(-\frac{I_m}{\omega C} \cos \omega t \right)^2$$

$$W_g = \frac{1}{2} L I_m^2 \sin^2 \omega t + \frac{1}{2} C \cdot \frac{I_m^2}{\omega^2 C^2} \cos^2 \omega t$$

$$W_g = \frac{1}{2} L I_m^2 \sin^2 \omega t + \frac{1}{2} L I_m^2 \cos^2 \omega t$$

$$** \boxed{W_g = \frac{1}{2} L I_m^2} \rightarrow (3)$$

$$** \boxed{W_g = \frac{1}{2} C V_m^2} ; \text{ where } V_m = \frac{I_m}{\omega C}$$

$$(\because \omega^2 = 1/LC)$$

$$L = 1/\omega^2 C$$

we will get $\leftarrow$ Sub in eq (3)

therefore

$$Q = \frac{\frac{1}{2} L I_m^2}{\left(\frac{I_m}{\sqrt{2}} \right)^2 \cdot R \cdot \frac{1}{\omega}}$$

$$C: \text{ Power dissipation} = I^2 R = \left(\frac{I_m}{\sqrt{2}} \right)^2 \cdot R$$

$$\Rightarrow \boxed{Q = \frac{\omega L}{R}}$$

$$\text{where } \omega = 1/\sqrt{LC}$$

$$\Rightarrow \boxed{Q = \frac{1}{R} \sqrt{\frac{L}{C}}}$$

$$\boxed{Q = \frac{\omega L}{R} = \frac{X_L}{R} = \frac{X_C}{R} = \frac{V_L}{V_R} = \frac{V_L}{V} \quad (\because V_R = V)} = \frac{Q_L}{P}$$

$$\boxed{Q = \frac{V_L \text{ (or)} V_C}{V}} \quad (\because V_L = V_C)$$

$$Q = \frac{X_L}{R} \Rightarrow Q = \frac{I^2 X_L}{I^2 R} = \frac{Q_L}{P}$$

$$\therefore \boxed{Q = \frac{Q_L}{P}}$$

$$\boxed{Q = \frac{X_L \text{ (or)} X_C}{R}} \quad (\because X_L = X_C)$$

$$\boxed{Q = \frac{X_C}{R} = \frac{1}{\omega C R}}$$

Under resonant ckt

$$Q = \frac{V_L \text{ (or)} V_C}{V}$$

since $(V_L \text{ (or)} V_C) > V$

$$Q > 1$$

$$\frac{X_L}{R} > 1 \Rightarrow \boxed{X_L > R}$$

$$\frac{X_C}{R} > 1 \Rightarrow \boxed{X_C > R}$$

$$Q \propto \frac{1}{\text{Power dissipation } (I^2 R = P)}$$

That means based on quality factor the power losses will be changes. & ~~from other changes.~~

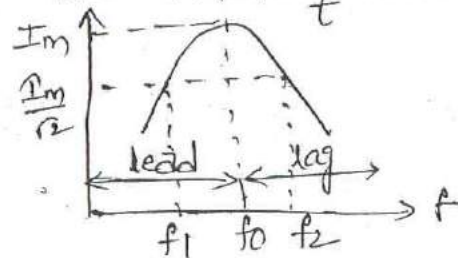
Bandwidth :-

* When the curve is drawn b/w Current & frequency then the curve is called as "resonance curve".

* Bandwidth is the range of frequencies on either side of the resonant frequency where the current falls from maximum value to 70.7% of the maximum value, and it is given by

$$\boxed{\text{Bandwidth} = f_2 - f_1}$$

f_2 = upper cutoff freq
 f_1 = lower cutoff freq



1) corresponds to $f_0 \rightarrow I = I_{\max}$
 $Z = R$

2) " " f_1 & $f_2 \rightarrow I = \frac{I_{\max}}{\sqrt{2}}$
 $Z = \sqrt{2} \cdot R$

(since $I \downarrow$ by $\sqrt{2}$ times $\Rightarrow Z \uparrow$ by $\sqrt{2}$ times)

3) corresponds to $f_0 \rightarrow \text{P.F. } \cos \theta = 1$

4) " " f_1 & $f_2 \rightarrow Z = R \pm jX$

$$Z = \sqrt{R^2 + X^2} = \sqrt{2} R$$

$$\Rightarrow R^2(2) = R^2 + X^2$$

$$\Rightarrow \boxed{X = R}$$

$$X = X_L - X_C$$

$$X_L = 2\pi f L ; X_C = \frac{1}{2\pi f C}$$

At f_1 :-
 $X_C > X_L$
 $X' = -ve$

At f_2 :- $X_L > X_C$
 $X = +ve$...

5) " " $f_1 \rightarrow Z = R - jX$

where $X = R$

$$\Rightarrow \text{impedance angle } \theta = \tan^{-1}\left(\frac{-X}{R}\right)$$

$$= -45^\circ$$

$$I = \frac{V \angle 0}{Z \angle -45} = \frac{V}{Z} \angle +45^\circ$$

$$P.F. \text{ angle} = +45^\circ$$

$$P.F. = \cos(45^\circ) = \frac{1}{\sqrt{2}} \text{ (lag)} \text{ since } X_L > X_C \text{ Inductive nature} \rightarrow \text{Capacitive nature}$$

Note:- Always Impedance angle & P.F. angle both are having equal magnitude with opposite sign.

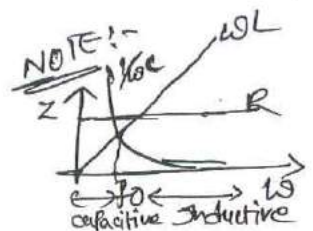
6) At f_1 :-

$$Z = R + jX$$

$$\text{where } X = R$$

$$\text{Impedance angle } \theta = \tan^{-1}\left(\frac{R}{R}\right) = +45^\circ$$

$$P.F. \text{ angle } \cos \theta = \cos(+45) = \frac{1}{\sqrt{2}} \text{ (lag)}$$



Note:- frequency < resonant freq $\Rightarrow$ capacitive nature > Inductive nature
 $\therefore P.F. = \text{lead}$

frequency > resonant freq $\Rightarrow$ Inductive nature > Capacitive nature
 $\therefore P.F. = \text{lag}$

7) corresponds to $f_1 \rightarrow X_C > X_L$; $X = R$

$$\text{Net react } X_C - X_L = R$$

$$\frac{1}{\omega_1 C} - \omega_1 L = R \rightarrow (1)$$

corresponds to $f_2 \rightarrow X_L > X_C$; $X = R$

$$X_{L2} - X_{C2} = R$$

$$\omega_2 L - \frac{1}{\omega_2 C} = R \rightarrow (2)$$

from Eq (1) & (2)

$$\frac{1}{\omega_1 C} - \omega_1 L = \omega_2 L - \frac{1}{\omega_2 C}$$

$$\Rightarrow \underline{\omega_1 \omega_2 = \frac{1}{LC}} \rightarrow (3)$$

$$\text{w.k.T } \omega_0^2 = \frac{1}{LC} \rightarrow (4)$$

from (3) & (4)

$$\omega_1 \omega_2 = \omega_0^2 \Rightarrow \omega_0 = \sqrt{\omega_1 \omega_2}$$

$$\Rightarrow \boxed{f_0 = \sqrt{f_1 f_2}}$$

Note:-
 $f_0 = \sqrt{f_1 f_2}$
 GENCO 2012

Add: eqn 1 & 2

$$\frac{1}{C} \left[\frac{1}{\omega_1} - \frac{1}{\omega_2} \right] + L [\omega_2 - \omega_1] = 2R$$

$$\frac{1}{C} \left[\frac{\omega_2 - \omega_1}{\omega_1 \omega_2} \right] + L [\omega_2 - \omega_1] = 2R$$

$$\frac{1}{C} \left[\frac{\omega_2 - \omega_1}{\omega_0^2} \right] + L [\omega_2 - \omega_1] = 2R$$

$$(\omega_2 - \omega_1)L + L [\omega_2 - \omega_1] = 2R$$

$$(\omega_2 - \omega_1) = \frac{2R}{2L} \text{ rad/sec}$$

$$f_2 - f_1 = \frac{1R}{\pi(2L)}$$

$$f_2 - f_1 = \frac{R}{2\pi L} \text{ Hz}$$

that means

only by knowing the ckt elements we can find out the B.W.

8) $Q = \frac{\omega L}{R}$

This quality factor can be define only under the resonant condⁿ

$$\Rightarrow Q = \frac{\omega_0 L}{R} = \frac{\omega_0}{R/L}$$

$$Q = \frac{\omega_0}{(\omega_2 - \omega_1)} = \frac{f_0}{f_2 - f_1}$$

(OR)

$$B.W = \frac{f_0}{Q}$$

$$B.W = \frac{f_0}{\frac{\omega_0 L}{R}} = \frac{f_0}{\frac{2\pi f_0 L}{R}}$$

$$f_2 - f_1 = \frac{R}{2\pi L}$$

9) corresponds to $f_0 \rightarrow I = I_{max}$
 $P = I_{max}^2 R$

" " $f_1 \text{ \& } f_2 \rightarrow I = \frac{I_{max}}{\sqrt{2}}$

$$P' = \frac{I_{max}^2}{2} R$$

$$P' = \frac{P}{2}$$

when compared to resⁿ condⁿ at cutoff frequ power becomes half.

Note :-

$$f_2 - f_1 = \frac{R}{2\pi L}$$

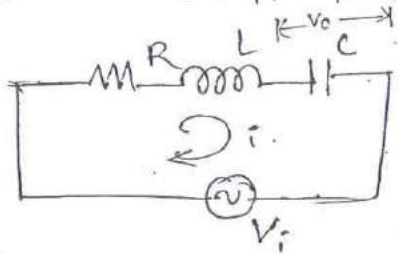
$$f_0 - f_1 = R/4\pi L$$

$$f_2 - f_0 = R/4\pi L$$

$$\omega_2 - \omega_1 = \frac{R}{L} \text{ rad/sec}$$

$$f_2 - f_1 = \frac{R}{2\pi L} \text{ Hz}$$

so the cutoff freq are called as "Half power frequencies".



By KVL

$$v_i(t) = i(t)R + L \frac{di(t)}{dt} + \frac{1}{C} \int i(t) dt$$

Differentiate the above eqn

$$\frac{dV_i(t)}{dt} = R \frac{di(t)}{dt} + L \frac{d^2 i(t)}{dt^2} + \frac{1}{C} i(t)$$

$$\Rightarrow s V_i(s) = s R I(s) + s^2 L I(s) + \frac{I(s)}{C}$$

$$\frac{d^2 i}{dt^2} + \frac{R}{L} \frac{di}{dt} + \frac{i}{LC} = \frac{dV_i/dt}{L}$$

$$\left(s^2 + \frac{R}{L} s + \frac{1}{LC} \right) i = 0$$

$\downarrow$ $\omega_2 - \omega_1$ $\downarrow$ ω_0

$$Q = \frac{\omega_0}{\omega_2 - \omega_1}$$

(or)

$$\frac{V_i(s)}{I(s)} = \frac{R + s^2 L + \frac{1}{C}}{s}$$

$$\frac{I(s)}{V_i(s)} = \frac{s}{L \left(s^2 + \frac{R}{L} s + \frac{1}{LC} \right)}$$

$$V_o(s) = \frac{1}{Cs} I(s)$$

$$\Rightarrow \frac{V_o(s)}{V_i(s)} = \frac{k_c}{s^2 + \frac{R}{L} s + \frac{1}{LC}}$$

$$\Rightarrow \alpha = \frac{R}{2\sqrt{L/C}}$$

$$\omega_n = 1/\sqrt{LC}$$

Now compare the above eq with $s^2 + 2\alpha\omega_n s + \omega_n^2 = 0$

$$\Rightarrow 2\alpha\omega_n = \frac{R}{L}$$

$$\omega_n^2 = \frac{1}{LC}$$

$$\Rightarrow \alpha = \frac{R}{2} \sqrt{\frac{C}{L}}$$

$$\omega_n = 1/\sqrt{LC}$$

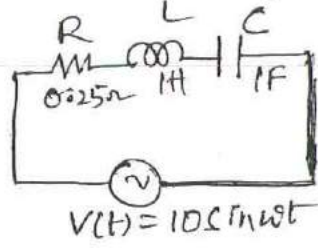
w.k.t $Q = \frac{1}{R} \sqrt{\frac{L}{C}}$

$$\Rightarrow \alpha = \frac{1}{2Q}$$

(pb)

sol:

find f_0 , Q , B.W, α , f_1 , f_2 , I at f_0 ?



$$1) f_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{1 \times 1}} = \frac{1}{2\pi} \text{ Hz}$$

$$2) Q = \frac{1}{R} \sqrt{\frac{L}{C}} = \frac{1}{0.25} \sqrt{\frac{1}{1}} = \frac{1}{0.25} = 4$$

$$3) \text{B.W} = \frac{R}{2\pi L} = \frac{0.25}{2\pi \times 1} = \frac{1}{8\pi}$$

$$4) \alpha = \frac{1}{2Q} = \frac{1}{2(4)} = \frac{1}{8}$$

$$5) f_2 - f_1 = \frac{1}{8\pi}$$

$$\sqrt{f_2 f_1} = f_0 \Rightarrow f_2 f_1 = \frac{1}{4\pi^2}$$

$$(f_1 + f_2)^2 - 4f_1 f_2 = (f_2 - f_1)^2$$

$$f_1 + f_2 = \left(\frac{1}{8\pi}\right)^2 + 4 \frac{1}{4\pi^2}$$

$$f_1 + f_2 = \frac{1}{64\pi^2} + \frac{1}{\pi^2} \Rightarrow$$

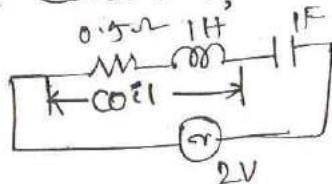
$$\Rightarrow f_1 + f_2 = \underline{\quad}$$

$$\therefore \begin{cases} f_1 - f_2 = \underline{\quad} \\ f_1 + f_2 = \underline{\quad} \end{cases} \Rightarrow \begin{cases} f_1 = \underline{\quad} \text{ Hz} \\ f_2 = \underline{\quad} \text{ Hz} \end{cases}$$

6) $I = \frac{V}{Z} = \frac{V}{R} = \frac{(10/12)}{0.25}$

$$\Rightarrow I = \frac{40}{12}$$

(Pb) Find voltage across of the coil of the ckt shown at resonance condition?



Since coil is nothing but combination of Resistor & Inductance.

Sol:-

$$Q = \frac{V_L}{V}$$

$$\Rightarrow Q = \frac{1}{R} \sqrt{\frac{L}{C}} = \frac{1}{0.5} \sqrt{\frac{1}{1}} = \underline{2}$$

$$\therefore V_L = \underline{4V}$$

(OR)

$$V_R = V$$

$$I \cdot R = 2$$

$$I \times 0.5 = 2$$

$$\Rightarrow I = \underline{4A}$$

$$\omega = \frac{1}{\sqrt{LC}} = 1 \text{ rad/sec}$$

$$X_L = \omega L$$

$$X_L = 1 \times 1 = 1$$

$$\therefore \text{Voltage across coil} = \sqrt{V_R^2 + V_L^2} = \sqrt{2^2 + 4^2} = \underline{\sqrt{20}}$$

$$V_L = I X_L = 4 \times 1 = \underline{4V}$$

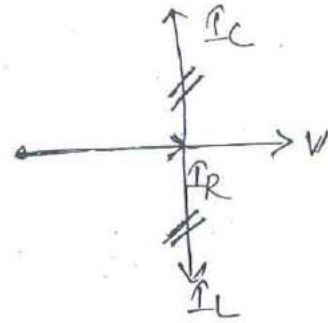
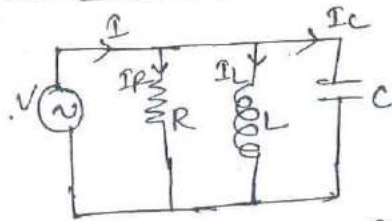
$$\text{Voltage across coil} = \sqrt{V_R^2 + V_L^2}$$

$$= \sqrt{2^2 + 4^2}$$

$$= \underline{\sqrt{20}}$$

Parallel Resonance :-

Case (i) :-



$$\text{By KCL: } I = I_R \angle 0 + I_L \angle -90 + I_C \angle 90$$

to satisfy defn

$$I = I_R \angle 0 + I_L \angle -90 + I_C \angle 90$$

$$\Rightarrow \boxed{I = I_R}$$

to satisfy defn

$$I_C = I_L$$

$$\frac{V}{X_C} = \frac{V}{X_L} \Rightarrow \boxed{X_L = X_C}$$

$$\Rightarrow \boxed{B_C = B_L}$$

$$\omega_0 C = \frac{1}{\omega_0 L}$$

$$\Rightarrow \boxed{\omega_0 = \frac{1}{\sqrt{LC}}} \text{ rad/sec}$$

$$\boxed{f_0 = \frac{1}{2\pi\sqrt{LC}}} \text{ Hz}$$

1. corresponding to Parallel ckt

$$Y = G + j(B_C - B_L)$$

$$Y_{\min} = G \Rightarrow Y_{\min} = G$$

2. $Z_{\max} = \frac{1}{Y_{\min}}$

3. $I_{\min} = \frac{V}{Z_{\max}}$

4. $\cos \theta = 1$

5. $I_R = I$

6. Net reactive current is equal to zero.

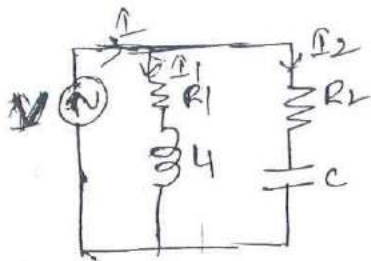
7. Current flowing through inductor (or) Current flowing through capacitor is greater than Total Current. This phenomena is called as "Current Magnification".

8. Parallel Resonant ckt is also called as "Anti Resonant ckt" (since all results are inversely proportional to series res)

9. Parallel Resonance ckt is also called as "Reflector ckt".

10. series Resonance ckt is called as "Acceptor ckt".

Case (ii) :-

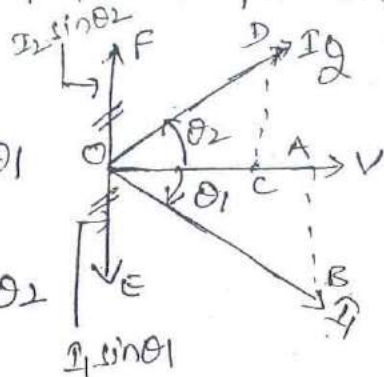


$$OA = I_1 \cos \theta_1$$

$$OE = AB = I_1 \sin \theta_1$$

$$OC = I_2 \cos \theta_2$$

$$OF = CD = I_2 \sin \theta_2$$



under Resonance Condⁿ net reactⁿ Current = 0.

$\Rightarrow$

$$B_L = B_C$$

$$\frac{X_L}{R_1^2 + X_L^2} = \frac{X_C}{R_2^2 + X_C^2}$$

$$\frac{\omega L}{R_1^2 + (\omega L)^2} = \frac{1/\omega C}{R_2^2 + (1/\omega C)^2}$$

$$I = I_1 \cos \theta_1 = I_2 \cos \theta_2$$

$\Rightarrow$ By rearranging the eqⁿ

$$\Rightarrow f_0 = \frac{1}{2\pi\sqrt{LC}} \sqrt{\frac{R_1^2 - L/C}{R_2^2 - L/C}}$$

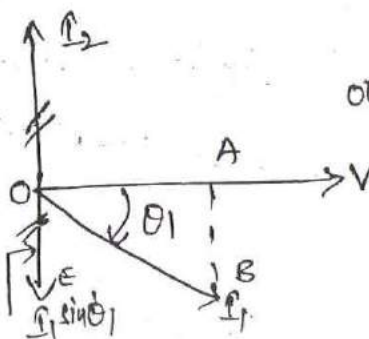
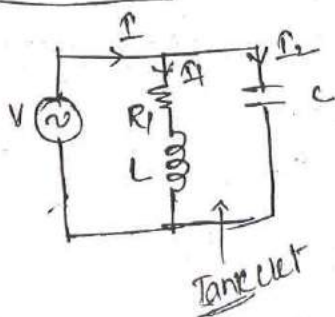
Total Current under resonance condition

$$I = VY$$

$$I = V \left[(G_1 + G_2) + j(B_C - B_L) \right]$$

$$I = V \left[\frac{R_1}{R_1^2 + X_L^2} + \frac{R_2}{R_2^2 + X_C^2} \right]$$

Case (iii) :-

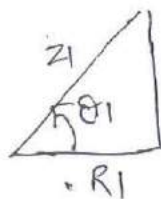


$$OA = I_1 \cos \theta_1$$

$$OE = AB = I_1 \sin \theta_1$$

$\approx$ under resⁿ Condⁿ

$$I = I_1 \cos \theta_1$$



$$\cos \theta_1 = \frac{R_1}{Z_1}$$

$$\sin \theta_1 = \frac{X_L}{Z_1}$$

under resonance condition

$$I_2 = I_1 \sin \theta_1$$

$$\frac{V}{X_C} = \frac{V}{Z_1} \sin \theta_1$$

$$\frac{V}{X_C} = \frac{V}{Z_1} \cdot \frac{X_L}{Z_1}$$

$$Z_1^2 = X_L X_C$$

$$Z_1 = \sqrt{X_L X_C}$$

$$= \sqrt{\omega L \cdot \frac{1}{\omega C}}$$

$$Z_1 = \sqrt{\frac{L}{C}} //$$

Note:- $Z_1 = \sqrt{\frac{L}{C}}$ $\rightarrow$ units = Ω

By knowing only circuit elements we can find out the Z_1 .

under resonance condition

$$I = I_1 \cos \theta_1$$

$$I = I_1 \cos \theta_1$$

$$I = \frac{V}{Z_1} \cos \theta_1$$

$$I = \frac{V}{Z_1} \cdot \frac{R_1}{Z_1}$$

$$I = \frac{V R_1}{Z_1^2}$$

$$I = \frac{V (40/R_1)}{R_1} \Rightarrow Z_{DY} = \frac{L}{R_1 C}$$

under resonance condition

$$B_1 = B_C$$

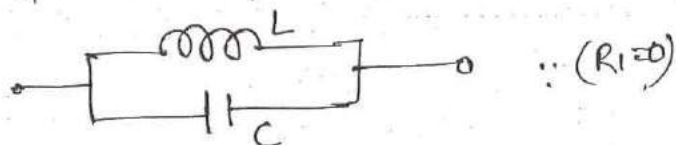
$$Z_1^2 = R_1^2 + X_L^2$$

$$\frac{L}{C} = R_1^2 + (\omega \pi f_0 L)^2$$

$$\Rightarrow f_0 = \frac{1}{2\pi} \sqrt{\frac{1}{LC} - \left(\frac{R_1}{L}\right)^2} //$$

This is the resonance frequency corresponding to tank circuit.

Note:- The given tank circuit is a practical tank circuit since internal resistance is considered, while corresponding to ideal tank circuit only L & C are present i.e. $R_1 = 0$.



$$\text{Now } Z_{DY} = \frac{L}{R_1 C} = \frac{L}{0 \cdot C} = \infty //$$

corresponding to Ideal Tank circuit.

$$f_0 = \frac{1}{2\pi \sqrt{LC}} //$$

Note:- In the tank circuit, the L & C continuous transformation of energy is nature. This impedance is called "dynamic impedance".

occurs oscillation is called

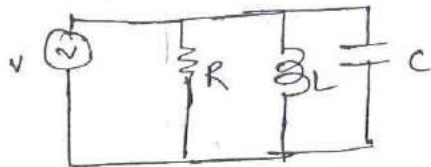
Q-Factor:-

$$Q = \frac{V_L \text{ (or)} V_C}{R} \rightarrow \text{series resonant ckt}$$

$$Q = \frac{I_L \text{ (or)} I_C}{I} \rightarrow \text{parallel resonant ckt}$$

this General Expt
is valid for any
combination
parallel ckt
↓
**

case (i):-



$$Q = \frac{I_L}{I} = \frac{I_L}{I_R}$$

Reactive Component of Current
Active Compⁿ of Current

$$Q = \frac{I_C}{I} = \frac{\frac{V}{X_C}}{\frac{V}{R}} = \frac{R}{X_C} = R\omega C$$

$$Q = \frac{I_L}{I_R} = \frac{\frac{V}{X_L}}{\frac{V}{R}} = \frac{R}{X_L} = \frac{R}{\omega L}$$

$$(\because \omega = \frac{1}{\sqrt{LC}})$$

$$\Rightarrow (Q = R\omega L = R\sqrt{\frac{L}{C}})$$

$$Q = R\sqrt{C/L}$$

$$Q = \frac{1/X_L}{1/R} = \frac{B_L}{G}$$

$$Q = \frac{B_L \text{ (or)} B_C}{G} \quad (\because B_L = B_C)$$

Parallel resonance ckt is a Current Magnifying ckt i.e.

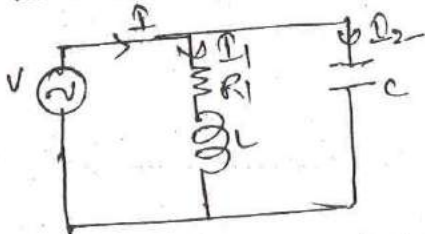
$$(I_L = I_C) > I$$

$$\Rightarrow Q > 1$$

$$\frac{R}{X_L} > 1 \Rightarrow R > X_L$$

$$\frac{R}{X_C} > 1 \Rightarrow R > X_C$$

Tank ckt:-



$$Q = \frac{\text{Reaⁿ Compⁿ of Current}}{\text{Active " "}}$$

$$= \frac{I_1 \sin \theta_1 \text{ (or)} I_2}{I}$$

under resⁿ cond

I_T in phase with $V \Rightarrow$ i.e. Active compⁿ

$$Q = \frac{I_2}{I} = \frac{\frac{V}{X_C}}{\frac{V}{\left(\frac{L}{R}\right)C}}$$

note: for tank ckt under resonance condⁿ

$$\frac{V}{I} = \frac{L}{R/C}$$

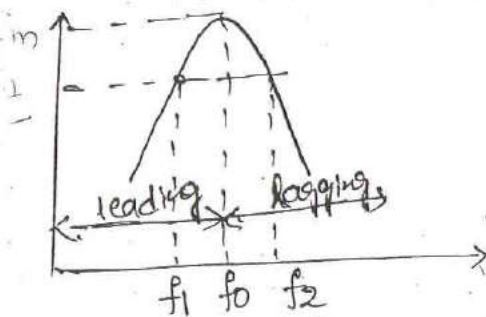
$$Q = \frac{\frac{R/C}{L \times X_C}}{\frac{L}{R/C}} = \frac{R/C}{L} \times \frac{L}{R/C}$$

$$Q = \frac{\omega L}{R} = \frac{X_L}{R}$$

for case (ii) $Q = \frac{\omega L}{R}$; for case (i) $Q = \frac{R}{\omega L}$ i.e. result is opposite
but general expression is same...

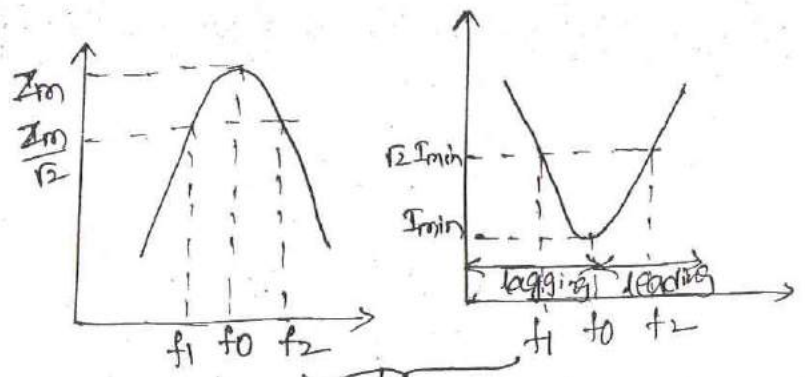
note:-
out of all tank ckt results are important.

$$Q > 1 \Rightarrow X_L > R$$



series

$$BW = f_2 - f_1$$



Parallel

$$BW = f_2 - f_1$$

$$BW = f_2 - f_1$$

$$B_L = \frac{1}{2\pi f L}; B_C = 2\pi f C$$

At low freqⁿ :- $B_L > B_C$

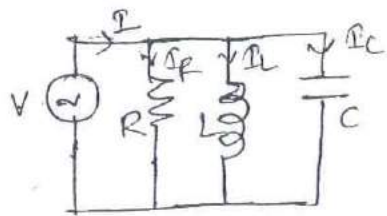
Inductive nature > capacitive nature

$\Rightarrow$ lag P.F

At high freqⁿ :- $B_L < B_C$

Indⁿ < capⁿ nature

$\Rightarrow$ lead P.F



note:-
(corresponding to parallel ckt the 'D' coeff indicates B.W = $1/RC$ like series RLC ckt.)

By KCL:-

$$I = \frac{V}{R} + \frac{1}{L} \int V dt + C \frac{dV}{dt}$$

Differentiate w.r.t to t .

$$\frac{dI}{dt} = \frac{1}{R} \frac{dV}{dt} + \frac{1}{L} \cdot V + C \frac{d^2V}{dt^2}$$

$$\Rightarrow \frac{d^2V}{dt^2} + \frac{1}{RC} \frac{dV}{dt} + \frac{V}{LC} = \left(\frac{dI}{dt}\right) \frac{1}{C}$$

$$\left(D^2 + \frac{1}{RC} D + \frac{1}{LC}\right) V = 0$$

B.W $\omega_2 - \omega_1 = 1/RC$ $\omega_0 = 1/LC$

$$Q = \frac{\omega_0}{\omega_2 - \omega_1}$$

$$\text{Now } s^2 + 2\alpha\omega_n s + \omega_n^2 = 0$$

$$\Rightarrow \alpha = 1/RC ; 2\alpha\omega_n = \frac{1}{RC}$$

$$\alpha = \frac{1}{2RC}$$

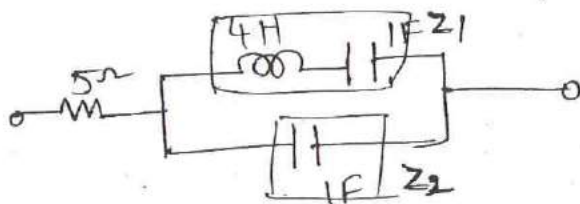
Now 'Q' only in terms of ckt elements is given by

$$Q = R \sqrt{\frac{C}{L}}$$

$$\Rightarrow Q = \frac{1}{2\alpha R}$$

$$Q = \frac{1}{2R} \sqrt{\frac{L}{C}}$$

(pb) find f_0 for the ckt shown?



sol:-

TO find resonant freq for any combination of the N/W

(i) find eqn impedance

(ii) equate imaginary part of the impedance to zero.

$$Z_1 = j(X_L - X_C)$$

$$Z_1 = j(\omega L - \frac{1}{\omega C})$$

$$= j(4\omega - \frac{1}{\omega})$$

$$Z_2 = -jX_{C2} = -\frac{j}{\omega}$$

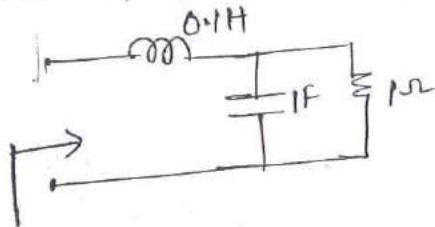
$$Z'_{eq} = \frac{Z_1 Z_2}{Z_1 + Z_2} = \frac{j(4\omega - \frac{1}{\omega}) (\frac{-j}{\omega})}{j(4\omega - \frac{1}{\omega}) - j/\omega} \times \frac{j}{j}$$

$$\text{Im}(Z'_{eq}) = 0 \Rightarrow (4\omega - \frac{1}{\omega})(\frac{1}{\omega}) = 0$$

$$\Rightarrow \omega = 0.5 \text{ rad/sec}$$

(pb)

find f_0 of the ckt shown?



$$\text{Now } \text{Im}(Z_{eq}) = 0$$

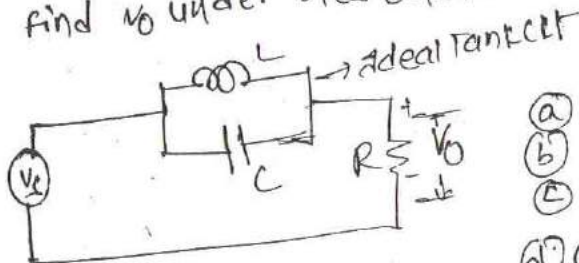
$$0.1\omega \times 1 = 0$$

$$\Rightarrow \omega = 10 \text{ rad/sec}$$

$$\Rightarrow 0.1\omega^2 - 0.1 - 1 = 0 \Rightarrow \omega^2 - 1 - 10 = 0 \Rightarrow \omega^2 = 9 \Rightarrow \omega = 3 \text{ rad/sec}$$

(pb)

find V_0 under resonance cond

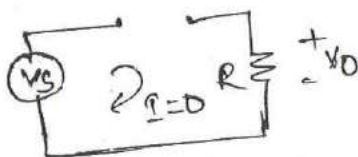


- (a) V_s
- (b) $V_s/2$
- (c) $V_s/3$
- (d) 0

Sol:-

$$\text{for Ideal tank ckt } Z_{dy} = \infty$$

$\Rightarrow$ open that branch.



$$I = 0 \Rightarrow V_0 = I \cdot R = 0$$

(pb)

A parallel RLC ckt have effective resistance of the ckt?

$C = 0.1 \mu F$; $BW = 1 \text{ kHz}$. find

Sol:-

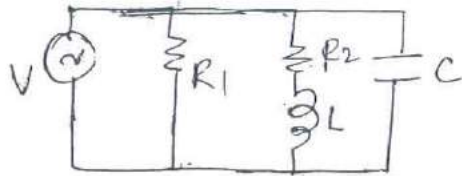
$$B.W = \frac{1}{RC} \text{ rad/sec}$$

$$\Rightarrow f_2 - f_1 = \frac{1}{2\pi RC} \text{ Hz}$$

$$\Rightarrow 1 \times 10^3 = \frac{1}{2 \times 3.14 \times 0.1 \times 10^{-6} \times R}$$

$$\Rightarrow R = \underline{\underline{159.15 \Omega}}$$

Q6



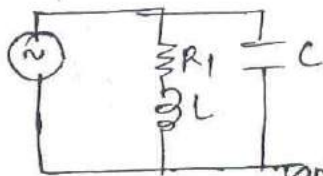
when $R_1 \uparrow \Rightarrow \phi ?$

$R_2 \uparrow \Rightarrow \phi ?$

Sol:-



$$\Rightarrow \phi = \frac{R_1}{\omega L}$$



$$\Rightarrow \phi = \frac{\omega L}{R_1}$$

Tan delta

In the problem if only R_1 is present in any branch, so we use 1st formula. while for R_2 we use 2nd formula.

As $R_1 \uparrow \Rightarrow \phi \uparrow$

As $R_2 \uparrow \Rightarrow \phi \downarrow$

.....

 x